incidence, . From ,

6.

The equation

is even in ; the ray is symmetric about the polar axis. Left of the origin, ; on the right, . The figure shows , with axis and dwarf at . In , insert the periapsis coordinates and note , giving

Rays past both sides of the dwarf subtend at the observer; this can be expressed in different ways, yielding different solutions.

Method 1: let be the dwarf–Earth distance. Approximately,

In , substitute observer coordinates :

Eliminate from and to find

Substitute into :

Thus

Here . Numerical substitution gives

Method 2: coordinates of the ray going to infinity can be written

Take , insert into , set the denominator to zero; note :

Thus

Use and note . Equations and agree, again giving